

Cloud-based machine learning can train predictive models

Machine learning is the use and development of algorithms and statistical models to teach computer systems to analyze and discover patterns in data. When you develop a machine learning model, it's important to have access to both enough data, and the correct data. As a data analyst, you may use machine learning to train models to analyze and predict using your data. In this reading, you'll learn more about how cloud-based computing services are used to train and use machine models.

The benefits of cloud-based machine learning

Building machine learning models requires owning both processing power and data storage. Traditionally, these resources require physical machines to host data and process it to create a model. This means that the machine learning model exists on a single machine, or a networked group of machines. Cloud computing enables you to host your data, and use cloud infrastructure to build a machine learning model in an environment that is both accessible to your team, and far more scalable.

Running machine learning models in the cloud can also save on overall costs. You can use a pay-per-use model, or select less expensive storage tiers for your data. And while many cloud providers have their own machine learning services, training models in the cloud may require less technical expertise.

How cloud-based machine learning trains models

To get started, you'll first need to prepare your data. For example, consider a local university that wants to predict student final grades using a machine learning model. During data preparation, they'll need to ensure that the data is clean, and that they can create a summary. A clean dataset means that there's no data missing, and no duplicate entries. Clean data will generate the most accurate machine learning model possible. And the summary of that data will create a foundation to train the model.

Next, they'll need to choose a training method. Methods could include training without code, writing unique code, or different algorithms depending on what kind of information they're trying to predict. At this stage, the cloud-based processing services will perform the majority of the work, including analyzing the existing test scores, and developing the model through model training.

Then, the university's academic data team will need to evaluate the model and work through iterations to ensure it's accurate. The academic data team may need to make adjustments to the data based on outliers or special circumstances, like a change in textbooks. If they use model evaluation metrics, they can determine whether the model is working by assessing how accurate the model's predictions are from the test data.

After they evaluate the model and confirm that it's accurate, they can deploy the model to production. At this stage, the machine learning model will start to create predictions on new data based on the existing historical data. The university can then create a prediction for final grades for the next semester.

You can use cloud services to create near real-time predictions, or batch predictions when you don't need synchronous predictions, or need to predict several outcomes at once.

Finally, the university's academic data team will monitor the model to ensure it's available, and that the results are accurate. Cloud-based models can be configured to provide near real-time alerts for any setting they implement. They can also be retrained with new data over time to maintain and improve accuracy.

Key takeaways

Machine learning provides a way to automatically generate patterns using existing and ingested data. Because of the workload required to analyze data in this way, cloud-based machine learning models can provide an efficient way to generate models using existing models, and open source algorithms, along with scalable compute and data storage.

Resources for more information

To learn more about training machine learning models in the cloud, check out this link:

- Google Cloud provides information about training machine learning models in its Vertex AI documentation:
<https://cloud.google.com/vertex-ai/docs/start/introduction-unified-platform>